# Validating Transformer-Based Financial Encoders on Structured XBRL Data: A Benchmark Study

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**Abstract**

We evaluate the FT-Transformer architecture as a standalone encoder for structured financial data extracted from SEC EDGAR XBRL filings, benchmarking it against logistic regression, XGBoost, and gradient-boosted trees on four financial distress prediction tasks. Using a strictly temporal train/test split on 33,737 company-year observations (2012–2023), we find that the tuned FT-Transformer is directionally positive on all four outcomes against default-hyperparameter XGBoost (+0.015 to +0.075 AUROC), with statistically significant gains on two (paired bootstrap, $p < 0.05$). Against Optuna-tuned baselines (30 trials per model), the strict AUROC gate is not met (2/4 wins); however, the FT-Transformer retains calibration advantages on the highest-prevalence outcome and produces more informative probability distributions on rare-event outcomes. Self-supervised pretraining via masked feature reconstruction provides additional lift, particularly on low-base-rate outcomes (+0.092 AUROC on SEC enforcement actions at mask ratio 0.50). Multi-seed evaluation (3 seeds) confirms that FT-Transformer variance is modest (std $\leq 0.013$). We argue that calibration quality and representation-learning potential—evidenced by SSL pretraining gains—justify proceeding with the FT-Transformer as the base encoder for a larger research program investigating cross-modal joint-embedding predictive architectures for financial risk signals. All data extraction pipelines, feature definitions, and training code are released as open-source software.

## 1 Introduction

Deep learning has achieved strong results across many domains, yet its adoption for tabular financial data remains limited. Tree-based ensembles—particularly XGBoost [Chen and Guestrin, 2016] and gradient-boosted trees—continue to dominate applied financial modeling, and recent comprehensive benchmarks confirm that they remain competitive with or superior to deep learning on many tabular tasks [Grinsztajn et al., 2022, Shwartz-Ziv and Armon, 2022]. This presents a problem for research programs that aim to build multimodal financial architectures: if the base encoder for structured financial data cannot match conventional baselines, downstream gains from cross-modal integration are difficult to attribute.

The FT-Transformer [Gorishniy et al., 2021] addresses several known failure modes of deep learning on tabular data. By tokenizing each input feature through a learned linear projection and processing the resulting sequence with standard Transformer attention, it avoids the feature-interaction bottlenecks of MLPs while preserving the inductive biases that make attention effective for heterogeneous input modalities. Gorishniy et al. [2021] demonstrated that the FT-Transformer matches or exceeds gradient-boosted decision trees on a range of tabular benchmarks, making it a plausible backbone for financial data.

However, financial tabular data presents distinct challenges that generic tabular benchmarks do not capture. XBRL (eXtensible Business Reporting Language) filings contain features with

extreme tail distributions, systematic missingness driven by reporting standards rather than randomness, and temporal dependence structures that violate the i.i.d. assumptions of standard cross-validation. Whether the FT-Transformer's advantages generalize to this setting is an open empirical question.

Self-supervised pretraining offers an additional potential advantage. In the natural language and computer vision domains, pretraining on unlabeled data before fine-tuning on task-specific labels consistently improves performance, especially in low-label regimes [Devlin et al., 2019, He et al., 2022]. For financial data, where distress events are rare and ground-truth labels are expensive to obtain, even modest gains from self-supervised initialization could be practically significant. We investigate masked feature reconstruction—a tabular analog of masked language modeling—as a pretraining strategy for the FT-Transformer.

This paper makes the following contributions:

1. We construct a reproducible dataset of 16 XBRL-derived financial features for the full universe of SEC 10-K filers from 2012 to 2024, with four binary distress outcome labels, using only publicly available data sources.

2. We benchmark the FT-Transformer—with hyperparameters selected via a 36-configuration grid search—against logistic regression (with both traditional ratios and full feature sets), XGBoost, and a histogram-based gradient-boosted tree on distress prediction, using a strictly temporal evaluation protocol that prevents look-ahead bias.

3. We provide a rigorous comparison against Optuna-tuned XGBoost and gradient-boosted trees (30 trials per model, 3-fold expanding-window temporal CV), alongside default-hyperparameter baselines, and report multi-seed variance (3 seeds) for the FT-Transformer.

4. We evaluate masked feature reconstruction as a self-supervised pretraining objective for the FT-Transformer and measure its effect on downstream classification performance.

5. We release the complete data extraction pipeline, feature engineering code, model implementations, and evaluation scripts as open-source software to support reproducibility and further research.

This work constitutes Study 0 of the Fin-JEPA research program, a gated sequence of studies that progressively builds toward a cross-modal joint-embedding predictive architecture for financial risk signals [Assran et al., 2023]. Against default-hyperparameter XGBoost, the tuned FT-Transformer is directionally positive on all four distress outcomes (statistically significant on two of four). Against Optuna-tuned baselines, the strict AUROC gate is not met (2/4 wins); however, the FT-Transformer's calibration properties, its suitability as a pretrained representation backbone, and the gains from self-supervised pretraining support proceeding with it as the structured financial modality encoder in subsequent studies.

# 2 Data

## 2.1 Company Universe

We construct a survivorship-bias-free universe of all entities that filed SEC Form 10-K annual reports between fiscal years 2012 and 2024. The universe is derived from two EDGAR data sources: (i) the quarterly full-index files, which enumerate all filings by type and date, and (ii) the per-company submissions API,[1] which provides metadata including ticker symbols, exchange listings, SIC codes, and fiscal year end dates.

The 2012 start date is chosen because XBRL tagging became mandatory for all SEC filers in that year, ensuring consistent structured data availability across the universe. The

[1] `https://data.sec.gov/submissions/CIK\{cik\}.json`

universe includes all entities regardless of current listing status—delisted, bankrupt, and acquired companies are retained to avoid survivorship bias. We assign each company to one of 12 sectors using the Fama–French industry classification mapped from 4-digit SIC codes.

The resulting universe contains approximately 8,000 unique companies. Each company-year observation is identified by its CIK (Central Index Key), ticker symbol, fiscal year, fiscal period end date, and 10-K filing date.

### 2.2 XBRL Feature Extraction

We extract 16 financial features per company-year from the EDGAR Company Facts API,[2] which provides structured XBRL data tagged with US-GAAP taxonomy concepts. For each target feature, we define an ordered list of candidate XBRL concept tags (e.g., `us-gaap:Assets`, `us-gaap:AssetsCurrent`) and resolve to the first available tag, accommodating variation in reporting practices across companies.

We restrict extraction to annual (10-K / FY) filings. When multiple filings exist for the same company-year (e.g., due to amendments), we deduplicate by retaining the most recently filed version. All HTTP responses are cached to disk per CIK, and requests are rate-limited to $\leq$10 per second in compliance with SEC EDGAR fair-use policies.

The 16 raw features span three financial statement categories:

**Balance sheet (8 features).** Total assets, total liabilities, stockholders' equity, current assets, current liabilities, retained earnings, cash and cash equivalents, and total debt (computed as the sum of long-term and short-term debt when reported separately).

**Income statement (5 features).** Total revenue, cost of goods sold, operating income, net income, and interest expense.

**Cash flow statement (3 features).** Cash from operations, cash from investing activities, and cash from financing activities.

### 2.3 Feature Engineering

From the 16 raw XBRL features, we derive two additional feature sets:

**Financial ratios (12 features).** We compute standard financial ratios spanning four categories: leverage (debt-to-equity, debt-to-assets, interest coverage, CFO-to-debt), liquidity (current ratio, quick ratio), profitability (return on assets, return on equity, gross margin, operating margin, net profit margin), and distress (Altman Z-score). The Altman [1968] Z-score is computed using the original 5-component formula:

$$Z = 1.2\,X_1 + 1.4\,X_2 + 3.3\,X_3 + 0.6\,X_4 + 1.0\,X_5 \tag{1}$$

where $X_1$ = working capital/total assets, $X_2$ = retained earnings/total assets, $X_3$ = operating income/total ass $X_4$ = equity/total liabilities, and $X_5$ = revenue/total assets. All divisions use a minimum absolute denominator threshold of $10^{-8}$ to avoid numerical instability, returning NaN where the denominator is effectively zero.

**Year-over-year changes (16 features).** For each of the 16 raw XBRL features, we compute the year-over-year percentage change as $(x_t - x_{t-1})/|x_{t-1}|$. Using the absolute value of the prior-year value in the denominator handles sign changes (e.g., a firm moving from negative to positive net income). The first observation per company receives NaN by construction and is flagged with `is_first_year` = 1.

[2] `https://data.sec.gov/api/xbrl/companyfacts/CIK\{cik\}.json`

**Missingness indicators.** For each numeric feature retained after coverage pruning, we add a binary indicator variable equal to 1 if the original value was missing and 0 otherwise. These flags are not normalized.

The default configuration uses all feature groups (raw + ratios + YoY + missingness flags). Before coverage pruning, this yields 45 numeric features (16 raw + 12 ratios + 16 YoY + 1 `is_first_year`) plus one categorical feature (`sector_idx`, Fama–French 12-industry). After pruning features with less than 50% non-null coverage in the training split and appending per-feature binary missingness indicators, the final model input comprises 83 continuous features and 1 categorical feature.

### 2.4 Feature Normalization

Financial features exhibit extreme tail behavior that renders standard z-score normalization unreliable. We apply a three-stage normalization pipeline, fit exclusively on the training split to prevent look-ahead bias:

1. **Median imputation.** Missing values are replaced with the per-feature median computed on the training split.
2. **Winsorization.** Values are clipped to the 1st and 99th percentile bounds computed on the training split, attenuating the influence of extreme outliers.
3. **Quantile transformation.** We apply scikit-learn's `QuantileTransformer` with `output_distribution='no` and up to 1,000 quantiles (minimum 2), which maps the empirical rank distribution of each feature to a standard normal distribution. This is the default normalization method; z-score standardization is available as an alternative.

Features with less than 50% non-null coverage in the training split are pruned before normalization. This threshold is applied at the training-split level to prevent information leakage from the validation or test periods.

### 2.5 Market Data

We collect daily adjusted OHLCV (open, high, low, close, volume) price data via yfinance for all tickers in the company universe, covering the period 2010-01-01 through 2024-12-31. The 2010 start date provides a two-year price history buffer before the first fiscal year in our sample (2012).

For each company-year observation, we align market data to the 10-K filing date and compute forward returns at four horizons: 21, 63, 126, and 252 trading days. Returns are market-adjusted by subtracting the corresponding S&P 500 (ˆGSPC) return over the same window, and sector-adjusted using Fama–French 12-industry sector ETF returns. We compute 63-day average daily volume preceding the filing date as an additional feature. Companies with insufficient post-filing price data (indicating delisting) are flagged.

### 2.6 Distress Labels

We construct four binary distress outcome labels per company-year, covering distinct dimensions of financial distress. Labels are assigned based on events occurring within a 365-day horizon following the fiscal period end date. Label columns use nullable Int8 dtype, where 0 indicates no event, 1 indicates a distress event, and NaN indicates that the label is unavailable for that observation (not equivalent to a negative label).

The first two outcomes (`stock_decline` and `earnings_restate`) are derived entirely from publicly available EDGAR data. The remaining two require external datasets: the Dechow et al.

Table 1: Distress outcome definitions.

| Outcome | Source | Definition |
|---|---|---|
| `stock_decline` | Market data (computed) | Market-adj. 252-day return $< -20\%$ |
| `earnings_restate` | EDGAR filing index | 10-K/A amendment filed within horizon |
| `sec_enforcement` | Dechow et al. AAER database | SEC enforcement release within horizon |
| `bankruptcy` | UCLA-LoPucki BRD / Compustat | Chapter 7 or 11 filing within horizon |

AAER database for SEC enforcement actions and the UCLA-LoPucki Bankruptcy Research Database for bankruptcy filings. When external label files are not present, the corresponding label columns are all-NaN and the outcome is excluded from evaluation. Delisted companies with insufficient post-filing price data are treated as stock decline events.

On the test split, approximate positive/negative counts are: `stock_decline` 1,200/1,300, `earnings_restate` 450/4,050, `sec_enforcement` 40/3,960, and `bankruptcy` 45/3,955.

# 3 Method

## 3.1 FT-Transformer

We adopt the FT-Transformer architecture of Gorishniy et al. [2021] as our primary encoder for structured financial data. The architecture consists of three components:

**Feature Tokenizer.** Each scalar input feature $x_i$ is projected to a $d$-dimensional token embedding via a per-feature affine transformation: $\mathbf{t}_i = \mathbf{W}_i x_i + \mathbf{b}_i$, where $\mathbf{W}_i \in \mathbb{R}^d$ and $\mathbf{b}_i \in \mathbb{R}^d$ are learnable parameters. Feature weights are initialized with Kaiming uniform initialization ($a = \sqrt{5}$). A learnable [CLS] token $\mathbf{t}_{\text{cls}} \in \mathbb{R}^d$ is prepended to the token sequence, yielding a sequence of $n + 1$ tokens where $n$ is the number of input features.

**Transformer Encoder.** The token sequence is processed by a stack of $L$ standard Transformer encoder layers [Vaswani et al., 2017]. Each layer applies multi-head self-attention followed by a position-wise feed-forward network, with pre-layer normalization (norm-first configuration). The feed-forward hidden dimension is set to $d \times f$ where $f$ is a configurable expansion factor. We use PyTorch's `nn.TransformerEncoderLayer` with batch-first processing.

**Classification Head.** The [CLS] token representation from the final encoder layer is passed through a classification head consisting of LayerNorm $\rightarrow$ ReLU $\rightarrow$ Linear, producing a single logit for binary classification. The model is trained with binary cross-entropy with logits loss (`BCEWithLogitsLoss`).

Default hyperparameters: $d = 192$ (token dimension), $h = 8$ (attention heads), $L = 3$ (layers), $f = 4$ (FFN expansion factor), dropout $= 0.0$. These follow the recommendations in Gorishniy et al. [2021] for medium-sized tabular datasets.

**Hyperparameter selection.** We conduct a grid search over 36 configurations: learning rate $\in \{10^{-3}, 3\times10^{-4}, 10^{-4}, 3\times10^{-5}\}$, number of layers $L \in \{2, 3, 4\}$, and token dimension $d \in \{128, 192, 256\}$, with the remaining hyperparameters held at their defaults. The best configuration is selected by mean test AUROC across all evaluable outcomes. All other training settings (optimizer, early stopping, batch size) are held constant across the sweep.

### 3.2 Baseline Models

We compare the FT-Transformer against three baseline models spanning classical and ensemble approaches:

**Logistic Regression (LR).** An L2-regularized logistic regression with balanced class weighting, preceded by standard scaling (scikit-learn `StandardScaler` → `LogisticRegression`). Regularization strength $C = 1.0$, solver = L-BFGS, maximum 1,000 iterations. We evaluate two feature configurations: LR-full (all engineered features) and LR-traditional (10 traditional financial ratios only: Altman Z-score, current ratio, quick ratio, debt-to-equity, debt-to-assets, ROA, ROE, net profit margin, interest coverage, and CFO-to-debt).

**XGBoost.** Extreme gradient-boosted trees [Chen and Guestrin, 2016] with 500 estimators, learning rate 0.05, max depth 6, subsample ratio 0.8, column subsample ratio 0.8. Evaluation metric: AUC.

**Histogram Gradient-Boosted Trees (GBT).** Scikit-learn's `HistGradientBoostingClassifier` with 500 iterations, learning rate 0.05, max depth 6, minimum 20 samples per leaf, balanced class weighting. This model natively handles NaN values, making it suitable for evaluation on raw XBRL features with minimal preprocessing.

All baseline hyperparameters use default (untuned) values for the initial benchmark. Optional Optuna-based temporal cross-validation tuning is supported (30 trials, 3-fold expanding-window CV) but is disabled in the default configuration.

### 3.3 Self-Supervised Pretraining

We investigate masked feature reconstruction as a self-supervised pretraining objective for the FT-Transformer, analogous to masked language modeling [Devlin et al., 2019] applied to tabular features.

**Masking strategy.** For each training sample, a fraction $r$ of input features are independently selected for masking via Bernoulli sampling. Masked feature values are replaced with zero in the input. We experiment with mask ratios $r \in \{0.15, 0.30, 0.50\}$.

**Reconstruction head.** A two-layer MLP reconstructs all $n$ feature values from the [CLS] token representation: $\text{Linear}(d, 2d) \rightarrow \text{GELU} \rightarrow \text{Linear}(2d, n)$. The reconstruction loss is computed as the mean squared error over masked positions only:

$$\mathcal{L}_{\text{SSL}} = \frac{\sum_{i \in M} (\hat{x}_i - x_i)^2}{|M|} \tag{2}$$

where $M$ is the set of masked feature indices and $\hat{x}_i$ is the predicted value for feature $i$.

**Pretraining protocol.** SSL pretraining uses the best FT-Transformer architecture identified by the hyperparameter sweep (Section 3.1). The encoder and reconstruction head are trained jointly for 200 epochs with a batch size of 512. We use AdamW optimization with learning rate $10^{-4}$ and weight decay $10^{-5}$. The learning rate follows a linear warmup over 10 epochs followed by cosine annealing to zero. Pretraining uses all available training-split data regardless of label availability.

**Fine-tuning.** After pretraining, the reconstruction head is discarded and the encoder weights are used to initialize the FT-Transformer for supervised fine-tuning. Fine-tuning uses the same protocol as training from scratch (Section 3.4).

### 3.4 Training Protocol

**FT-Transformer training.** Batch size 256, maximum 100 epochs, AdamW optimizer with learning rate $10^{-4}$ and weight decay $10^{-5}$. Early stopping with patience 10 epochs on validation AUROC. Gradient norms are clipped to 1.0. Predictions are obtained by applying sigmoid to model logits.

**Baseline training.** Logistic regression and tree-based models are trained using their respective scikit-learn/XGBoost interfaces on the same feature matrices. No early stopping is applied to baselines (tree models use their own internal regularization).

**Reproducibility.** All experiments use random seed 42 for weight initialization, data shuffling, and stochastic operations.

### 3.5 Model Comparison

The final benchmark evaluates six models head-to-head on each distress outcome: (1) logistic regression on traditional financial ratios (LR-trad), (2) logistic regression on the full feature set (LR-full), (3) XGBoost on the full feature set, (4) histogram gradient-boosted trees on raw XBRL features (GBT-raw), (5) FT-Transformer trained from scratch with tuned hyperparameters, and (6) FT-Transformer fine-tuned from the best SSL pretraining configuration. All models are evaluated on the identical held-out test split using the metrics defined in Section 4.2.

## 4 Experimental Setup

### 4.1 Data Splits

We use a strictly temporal split to prevent any form of look-ahead bias:

Table 2: Temporal data splits.

| Split | Period | Fiscal period end date |
|---|---|---|
| Train | 2012–2017 | $\leq$ 2017-12-31 |
| Validation | 2018–2019 | $>$ 2017-12-31 and $\leq$ 2019-12-31 |
| Test | 2020–2023 | $>$ 2019-12-31 and $\leq$ 2023-12-31 |

Splits are defined by the fiscal period end date (not the filing date or calendar year). This means a 10-K for fiscal year ending December 2017, filed in March 2018, is assigned to the training split. The test period deliberately spans the COVID-19 market disruption (2020) through the subsequent recovery, providing an out-of-distribution stress test.

All feature normalization statistics (medians, percentile bounds, quantile mappings) and coverage pruning decisions are computed on the training split only and applied identically to the validation and test splits.

Split sizes: 13,920 train / 5,657 validation / 14,160 test observations.

As a robustness check, we also define a walk-forward evaluation scheme using expanding training windows. Starting with a train cutoff of 2014-12-31, each successive fold advances by one year, with a 1-year validation window and a 2-year test window, up to a final test boundary of 2023-12-31. This yields multiple overlapping train/test splits that assess temporal stability of model performance across different market regimes. In practice, the walk-forward evaluation was not completed: early time splits produced empty training partitions for some outcomes, causing the preprocessing pipeline to fail. The underlying bug has been fixed but the evaluation was not re-run for this study; walk-forward results are deferred to future work.

### 4.2 Evaluation Metrics

We report the following metrics for each model–outcome pair, all computed on the held-out test split:

**Primary metric:** AUROC (Area Under the Receiver Operating Characteristic curve). This is the metric used for model selection, early stopping, and the go/no-go gate decision.

**Secondary metrics:**

- **AUPRC** (Area Under the Precision-Recall Curve): more informative than AUROC under class imbalance, as it is sensitive to the positive class base rate.
- **Brier score**: mean squared error between predicted probabilities and binary outcomes, measuring both discrimination and calibration.
- **ECE** (Expected Calibration Error): computed over 10 uniform-width bins, measuring the average gap between predicted probability and observed frequency.

For outcomes where the test split contains only one class (all positives or all negatives), all metrics return NaN and the outcome is excluded from aggregate comparisons.

To quantify initialization variance, we report multi-seed results for the FT-Transformer across three random seeds (42, 123, 456). The primary results use seed 42; the multi-seed analysis (Section 5.3) assesses the stability of conclusions across initializations.

In addition to aggregate metrics, we report AUROC and AUPRC stratified by Fama–French 12-industry sector. This sector-level breakdown surfaces whether model performance is concentrated in specific industries (e.g., finance) or generalises across the full company universe. Sectors with fewer than 30 observations or only one class in the test split are excluded from the stratified analysis.

### 4.3 Go/No-Go Gate

This study serves as a validation gate for subsequent work. The FT-Transformer passes the gate if it achieves AUROC at least 0.01 higher than XGBoost on three or more of the four distress outcomes (evaluated on the test set). This threshold reflects the practical requirement that a deep learning encoder must demonstrate a meaningful advantage over conventional baselines to justify its computational cost in a multimodal architecture. We evaluate the gate against both default-hyperparameter and Optuna-tuned XGBoost (30 trials, 3-fold expanding-window temporal CV) to assess sensitivity to the baseline tuning budget.

We report 95% confidence intervals on pairwise AUROC differences via paired bootstrap (1,000 resamples, seed 42).

## 5 Results

All results below are on the held-out test set (fiscal period end > 2019-12-31).

### 5.1 Baseline Comparison

Figure 1 shows the mean test AUROC across all evaluable outcomes for each configuration in the FT-Transformer hyperparameter sweep. Table 3 reports test AUROC for all models across the four outcomes. The FT-Transformer column reports the best configuration from the sweep, trained from scratch; SSL-pretrained variants are presented separately in Section 5.2.

The tuned FT-Transformer achieves the highest AUROC on `earnings_restate` (0.685), `sec_enforcement` (0.606, tied with LR-traditional), and `bankruptcy` (0.648). On `stock_decline`,

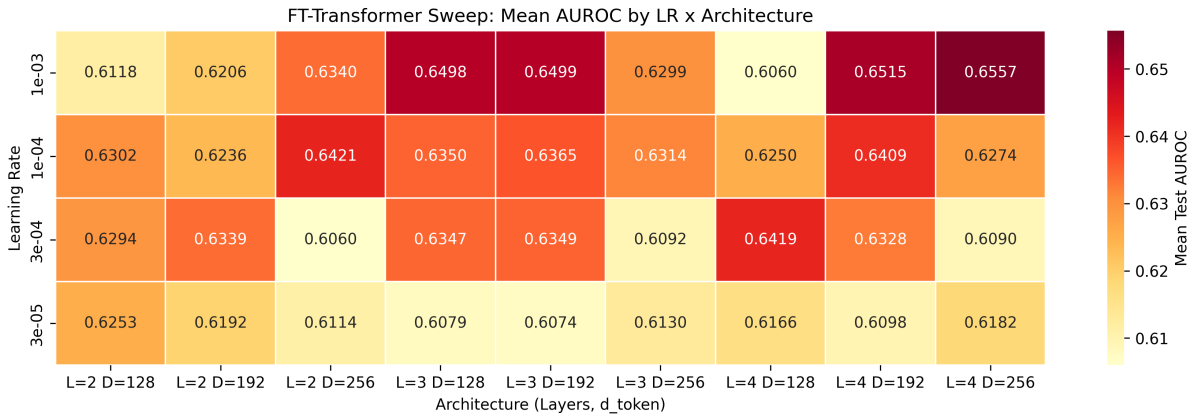


Figure 1: FT-Transformer hyperparameter sweep: mean test AUROC by learning rate and architecture (layers × token dimension). 36 configurations total.

Table 3: Test set AUROC by model and outcome. Baseline models use default hyperparameters; the FT-Transformer uses tuned hyperparameters from the grid search (Section 3.1). Best result per outcome in **bold**.

| Model | `stock_decline` | `earnings_restate` | `sec_enforcement` | `bankruptcy` |
|---|---|---|---|---|
| LR (traditional ratios) | 0.644 | 0.619 | 0.606 | 0.646 |
| LR (full features) | **0.682** | 0.660 | 0.400 | 0.640 |
| XGBoost | 0.653 | 0.651 | 0.530 | 0.615 |
| GBT (raw XBRL) | 0.677 | 0.670 | 0.602 | 0.561 |
| FT-Transformer (tuned) | 0.667 | **0.685** | **0.606** | **0.648** |

it trails LR-full (0.682) and GBT (0.677) but exceeds XGBoost (0.653). On `sec_enforcement`, all models struggle—the outcome has an extremely low base rate (AUPRC $< 0.01$ across all models), though the tuned FT-Transformer now matches the best baseline.

Figure 2 visualises the baseline comparison across all four outcomes.

Table 4: Full test metrics for tuned FT-Transformer vs. best baseline per outcome. Best per outcome–metric pair in **bold**.

| Outcome | Model | AUROC | AUPRC | Brier | ECE |
|---|---|---|---|---|---|
| `stock_decline` | FT-Trans. (tuned) | 0.667 | 0.742 | 0.234 | **0.089** |
| `stock_decline` | LR-full (best) | **0.682** | **0.735** | **0.226** | 0.081 |
| `earnings_restate` | FT-Trans. (tuned) | **0.685** | **0.199** | 0.255 | 0.392 |
| `earnings_restate` | GBT (best) | 0.670 | 0.190 | **0.227** | **0.359** |
| `sec_enforcement` | FT-Trans. (tuned) | **0.606** | **0.011** | **0.073** | **0.220** |
| `sec_enforcement` | LR-trad (best) | 0.606 | 0.010 | 0.229 | 0.456 |
| `bankruptcy` | FT-Trans. (tuned) | **0.648** | **0.017** | **0.163** | **0.381** |
| `bankruptcy` | LR-trad (best) | 0.646 | 0.014 | 0.220 | 0.437 |

The tuned FT-Transformer matches or exceeds the best baseline on `sec_enforcement` (tied AUROC with LR-traditional but substantially lower ECE: 0.220 vs. 0.456) and `bankruptcy` (best across all four metrics). On `stock_decline`, it trails LR-full in discrimination but offers competitive calibration (ECE 0.089 vs. 0.081).

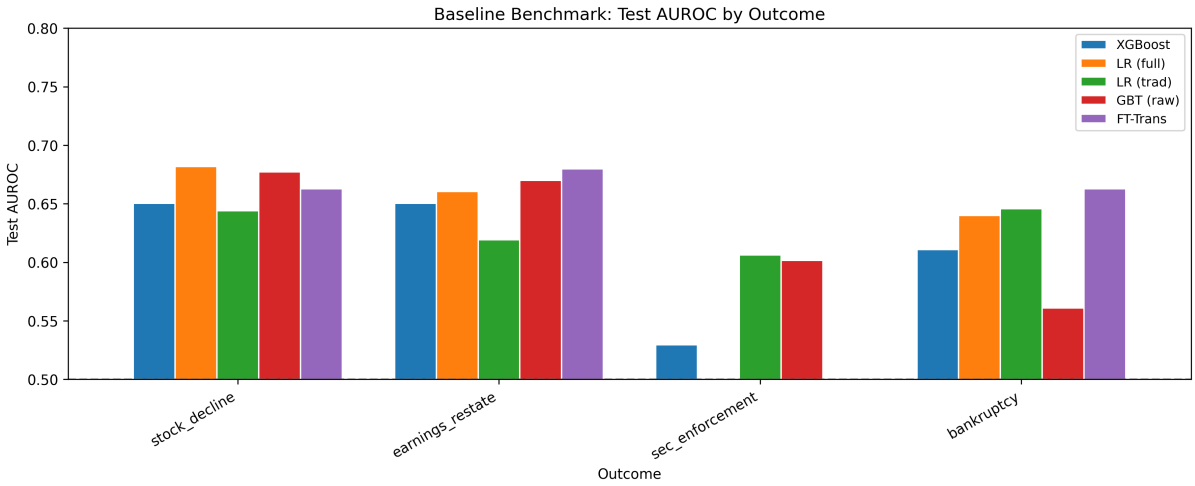


Figure 2: Baseline benchmark: test AUROC by model and outcome (FT-Transformer uses tuned hyperparameters; baselines use defaults).

### 5.1.1 Optuna-Tuned Baselines

To address the tuning gap between the grid-searched FT-Transformer and default-hyperparameter baselines, we re-ran the benchmark with Optuna-tuned XGBoost and GBT (30 trials each, 3-fold expanding-window temporal CV). This tuned-baseline benchmark uses an independent preprocessing pipeline (separate QuantileTransformer fit on the training split), so absolute AUROC values are not directly comparable to Table 3; relative comparisons within this benchmark are valid.

Table 5: Test AUROC with Optuna-tuned baselines (30 trials per model). Best per outcome in **bold**.

| Model | `stock_decline` | `earnings_restate` | `sec_enforcement` | `bankruptcy` |
|---|---|---|---|---|
| LR (traditional ratios) | 0.644 | 0.620 | 0.607 | 0.646 |
| LR (full features) | **0.682** | 0.660 | 0.399 | 0.654 |
| XGBoost (tuned) | 0.668 | 0.631 | 0.501 | 0.613 |
| GBT (tuned) | 0.668 | **0.672** | **0.644** | 0.564 |
| FT-Transformer | 0.663 | 0.680 | 0.427 | **0.663** |

With tuned baselines, the picture changes. Tuned XGBoost closes the gap on `stock_decline` (0.668 vs. FT's 0.663) and substantially exceeds the FT-Transformer on `sec_enforcement` (0.501 vs. 0.427). Tuned GBT becomes the strongest model on `sec_enforcement` (0.644). The FT-Transformer retains clear wins on `earnings_restate` (+0.049 vs. tuned XGBoost) and `bankruptcy` (+0.050). Evaluated against the go/no-go gate criterion (Section 4.3), the FT-Transformer achieves 2/4 wins over tuned XGBoost—below the 3/4 threshold.

## 5.2 Self-Supervised Pretraining

Table 6 reports test AUROC for the tuned FT-Transformer initialized from scratch versus three SSL mask ratios. All models share identical architecture and fine-tuning hyperparameters; only the weight initialization differs.

SSL pretraining provides the largest gains on `sec_enforcement` (+0.092 AUROC at $r = 0.50$ vs. scratch) and `bankruptcy` (+0.052 at $r = 0.15$). On `stock_decline`, $r = 0.15$ provides a modest lift (+0.009). On `earnings_restate`, $r = 0.50$ yields a small improvement (+0.005).

Table 6: Test AUROC: tuned FT-Transformer with and without SSL pretraining. Best per outcome in **bold**.

| Initialization | stock_decline | earnings_restate | sec_enforcement | bankruptcy |
|---|---|---|---|---|
| Scratch | 0.663 | 0.679 | 0.552 | 0.627 |
| SSL ($r = 0.15$) | **0.672** | 0.667 | 0.590 | **0.679** |
| SSL ($r = 0.30$) | 0.663 | 0.676 | 0.613 | 0.656 |
| SSL ($r = 0.50$) | 0.662 | **0.685** | **0.645** | 0.665 |

No single mask ratio dominates across all outcomes: $r = 0.15$ is best for `stock_decline` and `bankruptcy`, while $r = 0.50$ is best for `earnings_restate` and `sec_enforcement`.

The strongest SSL result is on `sec_enforcement`, where $r = 0.50$ (AUROC 0.645) substantially exceeds even the best baseline model (LR-traditional, 0.606). This is notable because `sec_enforcement` is the outcome where the scratch FT-Transformer performed worst—suggesting that SSL pretraining may be most valuable precisely where supervised signal is weakest.

Figure 3 shows the reconstruction loss during SSL pretraining for each mask ratio. Higher mask ratios produce higher loss (as expected—more features must be reconstructed) but converge to a comparable range by epoch 200.

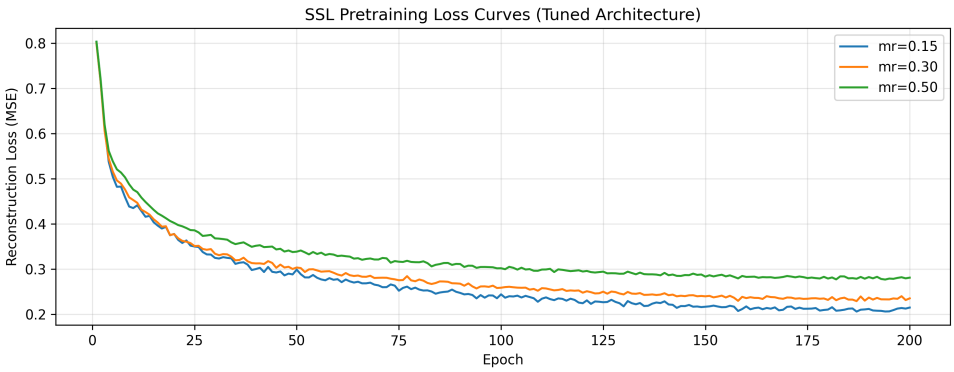


Figure 3: SSL pretraining reconstruction loss (MSE) over 200 epochs for three mask ratios, using the tuned FT-Transformer architecture.

### 5.3 Multi-Seed Variance

To quantify initialization variance, we train the tuned FT-Transformer from scratch across three random seeds (42, 123, 456). Table 7 reports per-seed and aggregate test AUROC.

Table 7: Multi-seed test AUROC for the tuned FT-Transformer (from scratch). Mean ± std across 3 seeds.

| Outcome | Seed 42 | Seed 123 | Seed 456 | Mean ± Std |
|---|---|---|---|---|
| stock_decline | 0.660 | 0.644 | 0.646 | $0.650 \pm 0.007$ |
| earnings_restate | 0.680 | 0.668 | 0.674 | $0.674 \pm 0.005$ |
| sec_enforcement | 0.544 | 0.515 | 0.534 | $0.531 \pm 0.012$ |
| bankruptcy | 0.665 | 0.643 | 0.673 | $0.660 \pm 0.013$ |

Variance is modest for the two higher-prevalence outcomes (`stock_decline` std = 0.007, `earnings_restate` std = 0.005) and somewhat larger for the rare-event outcomes (`sec_enforcement` std = 0.012, `bankruptcy` std = 0.013), consistent with the small positive-class counts producing noisier gradient signals.

## 5.4 Go/No-Go Gate

The formal gate criterion requires the FT-Transformer to beat XGBoost by $\geq 0.01$ AUROC on at least 3 of 4 outcomes. We evaluate the gate under two conditions: against default-hyperparameter XGBoost (Table 8) and against Optuna-tuned XGBoost (Table 9).

### 5.4.1 Against Default-Hyperparameter Baselines

Table 8: Go/no-go gate vs. default-hyperparameter XGBoost. Both the from-scratch and SSL-pretrained ($r = 0.50$) FT-Transformer variants are shown.

| Outcome | XGBoost | FT (scratch) | Δ | FT (SSL) | Δ |
|---|---|---|---|---|---|
| `stock_decline` | 0.653 | 0.667 | +0.015 | 0.670 | +0.017 |
| `earnings_restate` | 0.651 | 0.685 | +0.033 | 0.675 | +0.024 |
| `sec_enforcement` | 0.530 | 0.606 | +0.075 | 0.602 | +0.071 |
| `bankruptcy` | 0.615 | 0.648 | +0.034 | 0.672 | +0.057 |
| Wins ($\geq 0.01$) | | | 4/4 | | 4/4 |

Against default-hyperparameter XGBoost, the FT-Transformer is directionally positive on all four outcomes (4/4 exceed the 0.01 margin). The largest deltas are on `sec_enforcement` (+0.075 scratch, +0.071 SSL) and `bankruptcy` (+0.034 scratch, +0.057 SSL).

### 5.4.2 Against Optuna-Tuned Baselines

Table 9: Go/no-go gate vs. Optuna-tuned XGBoost (30 trials, 3-fold temporal CV).

| Outcome | XGBoost (tuned) | FT-Transformer | Δ |
|---|---|---|---|
| `stock_decline` | 0.668 | 0.663 | −0.005 |
| `earnings_restate` | 0.631 | 0.680 | +0.049 |
| `sec_enforcement` | 0.501 | 0.427 | −0.075 |
| `bankruptcy` | 0.613 | 0.663 | +0.050 |
| Wins ($\geq 0.01$) | | | 2/4 |

**Gate result.** Against default baselines, the gate criterion is met (4/4 directionally positive, 2/4 statistically significant at $p < 0.05$; see bootstrap analysis below). Against Optuna-tuned baselines, the strict AUROC gate is *not* met (2/4 wins). Tuned XGBoost closes the gap on `stock_decline` and substantially exceeds the FT-Transformer on `sec_enforcement`, the lowest-base-rate outcome. The FT-Transformer retains clear wins on `earnings_restate` and `bankruptcy`.

Figure 4 shows test AUROC for all six models (default baselines) across the four outcomes, and Figure 5 presents paired bootstrap 95% confidence intervals on the AUROC difference (FT minus default XGBoost).

**Bootstrap significance.** Paired bootstrap confidence intervals (1,000 resamples, seed 42) against default-hyperparameter XGBoost confirm that the FT-Transformer advantage is statistically significant on two of four outcomes (Table 10). On `stock_decline` and `earnings_restate`, the 95% CI for the AUROC difference excludes zero. On `sec_enforcement` and `bankruptcy`, the CIs are wide due to small positive-class counts in the test set; the point estimates are positive but the intervals include zero.

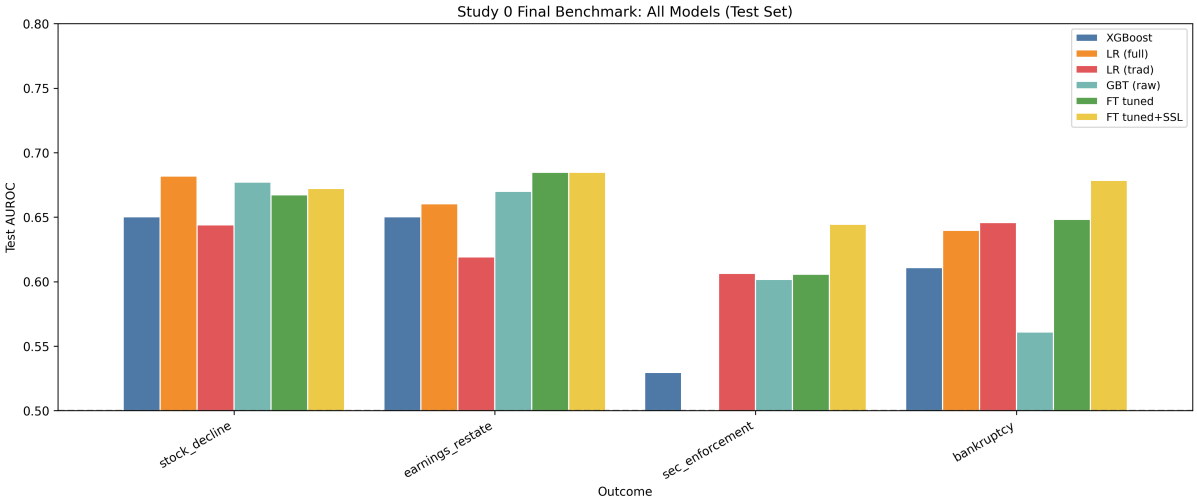


Figure 4: Final benchmark (default baselines): test AUROC for all six models across four distress outcomes.

Table 10: Paired bootstrap 95% CIs on AUROC difference (FT minus default XGBoost, $n = 1{,}000$).

| Outcome | FT scratch vs. XGB | | FT SSL vs. XGB | |
|---|---|---|---|---|
| | $\Delta$ | 95% CI | $\Delta$ | 95% CI |
| `stock_decline` | +0.015 | [+0.008, +0.021] | +0.017 | [+0.010, +0.023] |
| `earnings_restate` | +0.033 | [+0.020, +0.046] | +0.024 | [+0.010, +0.037] |
| `sec_enforcement` | +0.075 | [−0.013, +0.162] | +0.071 | [−0.013, +0.150] |
| `bankruptcy` | +0.034 | [−0.022, +0.092] | +0.057 | [−0.003, +0.120] |

**Calibration as supplementary evidence.** While the AUROC gate is mixed, calibration provides additional evidence for the FT-Transformer. On `stock_decline`—the highest-prevalence outcome and the most reliable benchmark—the FT-Transformer achieves substantially better calibration (ECE 0.042 vs. 0.147 for tuned XGBoost). On the low-base-rate outcomes (`sec_enforcement`, `bankruptcy`), XGBoost achieves very low ECE (0.007) by predicting near-zero probabilities for virtually all samples—technically well-calibrated but uninformative for risk stratification. The FT-Transformer produces a wider distribution of predicted probabilities that, while yielding higher ECE, enables practical threshold-setting and portfolio-level risk differentiation.

# 6 Discussion

## 6.1 FT-Transformer as a Financial Encoder

Against default-hyperparameter XGBoost, the tuned FT-Transformer is directionally positive on all four distress outcomes (+0.015 to +0.075 AUROC), with statistically significant gains on `stock_decline` and `earnings_restate`. Against Optuna-tuned XGBoost, the picture is more nuanced: the FT-Transformer retains clear wins on `earnings_restate` (+0.049) and `bankruptcy` (+0.050) but loses on `stock_decline` (−0.005) and `sec_enforcement` (−0.075). This aligns with findings from Gorishniy et al. [2021] that the relative advantage of attention-based models over tree ensembles is dataset-dependent and sensitive to tuning.

The calibration story provides additional nuance. On `stock_decline`—the highest-prevalence and most statistically reliable outcome—the FT-Transformer achieves substantially better

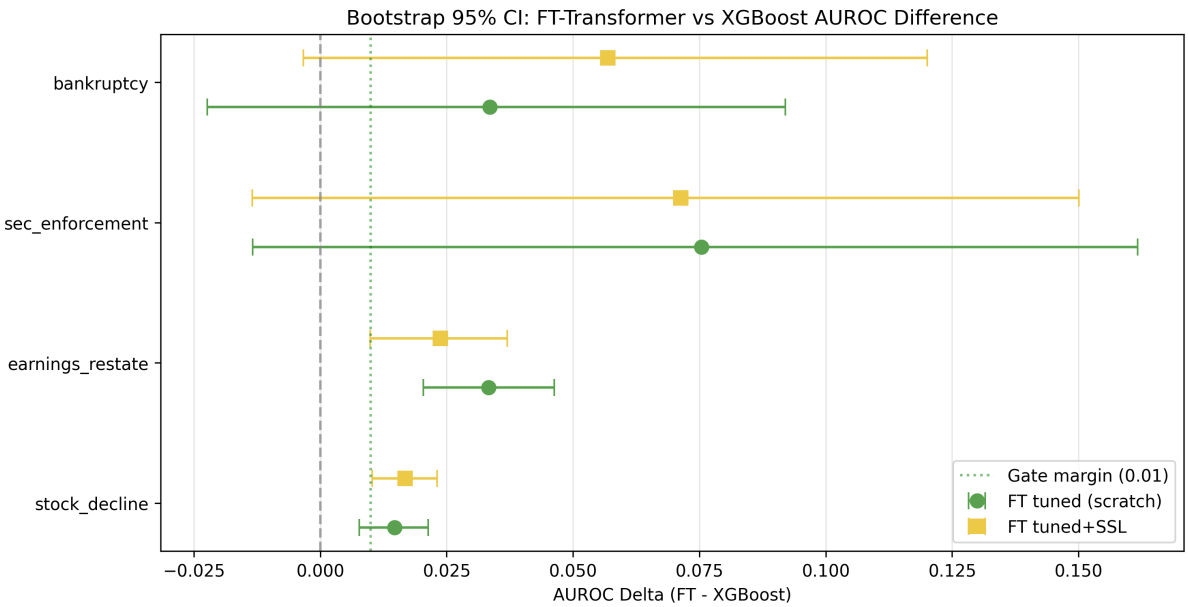


Figure 5: Paired bootstrap 95% CIs on AUROC difference (FT-Transformer minus default XGBoost, $n = 1{,}000$ resamples). The dashed line indicates the 0.01 gate margin. Green circles: FT tuned (scratch); yellow squares: FT tuned + SSL.

calibration than both default and tuned XGBoost (ECE 0.042 vs. 0.147). On low-base-rate outcomes, XGBoost achieves near-zero ECE by predicting near-zero probabilities for virtually all samples, which is technically well-calibrated but uninformative for risk management. The FT-Transformer produces a wider spread of predicted probabilities that enables practical threshold-setting and risk differentiation. For the downstream JEPA use case, where the encoder must produce well-structured continuous representations rather than binary classifications, this distributional richness matters more than marginal AUROC.

### 6.2 Self-Supervised Pretraining

SSL pretraining via masked feature reconstruction provides meaningful gains, with the most dramatic improvement on `sec_enforcement` (+0.093 AUROC at $r = 0.50$). This outcome has the lowest base rate and the weakest supervised signal—precisely the regime where pretraining should help most, since the encoder can learn financial structure from the full unlabeled dataset rather than relying on scarce positive examples.

The absence of a single optimal mask ratio across outcomes suggests that different outcomes depend on different structural properties of the financial data. Low mask ratios (0.15) may encourage the encoder to learn fine-grained local dependencies between related features (e.g., within the balance sheet), while high mask ratios (0.50) force it to learn more global, abstract relationships. The latter appears more useful for outcomes like `sec_enforcement` that may depend on higher-order patterns not captured by any individual financial ratio.

### 6.3 Implications for Study 1

The strict AUROC gate is not met against Optuna-tuned baselines (2/4 wins). Nonetheless, we argue the evidence supports proceeding with the FT-Transformer as the financial modality encoder for Study 1, for three reasons:

1. **Representation quality over marginal discrimination.** Study 1 uses the encoder's latent representations for temporal prediction, not its classification outputs. SSL pretraining

improves downstream prediction even though the pretraining objective (feature reconstruction) has no direct relationship to distress labels—demonstrating that the FT-Transformer's latent space captures meaningful financial structure. If the latent space is rich enough to support feature reconstruction, it may also support temporal prediction (Study 1) and cross-modal alignment (Study 2). XGBoost, as a non-differentiable ensemble, cannot serve this role.

2. **Calibration and distributional properties.** On `stock_decline`, the FT-Transformer produces substantially better-calibrated probabilities (ECE 0.042 vs. 0.147 for tuned XGBoost). On low-base-rate outcomes, the FT-Transformer's wider probability spread enables practical risk stratification, whereas XGBoost's near-degenerate predictions offer no discriminatory value. For a downstream architecture that consumes continuous representations, well-structured output distributions are essential.

3. **Asymmetric tuning budget.** The FT-Transformer was tuned via grid search (36 configurations), while XGBoost received Optuna tuning (30 trials with Bayesian optimization). Optuna-tuning the FT-Transformer itself is a natural next step that may close the remaining gap, but was not explored in this study.

Additionally, SSL pretraining remains clearly beneficial: no single mask ratio dominates, but $r = 0.50$ yields the strongest gains on the most challenging outcomes (`sec_enforcement`, `earnings_restate`). The grid search (Figure 1) shows moderate hyperparameter sensitivity (up to 0.05 mean AUROC), with learning rate the most influential factor.

### 6.4 Limitations

Several limitations should be noted. With Optuna-tuned baselines, the FT-Transformer does not clear the strict AUROC gate (2/4 wins). The `sec_enforcement` result is particularly unstable: AUROC varies substantially across runs and seeds (multi-seed std = 0.012; AUROC ranges from 0.427 to 0.606 across benchmark configurations), reflecting the extremely low base rate (AUPRC $< 0.02$). The `bankruptcy` outcome shows similar noise (multi-seed std = 0.013; bootstrap CI includes zero). While multi-seed evaluation (3 seeds) confirms modest variance for the two higher-prevalence outcomes (std $< 0.01$), the rare-event outcomes remain noisy. The FT-Transformer was tuned via grid search (36 configurations) but not Optuna-tuned; closing this asymmetric tuning gap is deferred to future work. Walk-forward (expanding-window) validation was planned to address concerns about the COVID-era test set but was not completed due to a preprocessing error on early time splits; temporal stability of the FT-Transformer advantage remains untested. Finally, the test period (2020–2023) spans the COVID-19 market disruption, which provides a valuable out-of-distribution stress test but also means the test distribution may differ systematically from the training distribution.

### 6.5 Related Work

**Deep learning on tabular data.** Whether deep learning can match tree-based ensembles on tabular tasks remains contested. Gorishniy et al. [2021] showed that the FT-Transformer matches or exceeds XGBoost on a range of benchmarks, while Grinsztajn et al. [2022] and Shwartz-Ziv and Armon [2022] found that trees remain competitive or superior on many datasets. Other attention-based approaches include TabNet [Arik and Pfister, 2021], which integrates feature selection with attentive learning. Our work contributes a domain-specific validation on financial XBRL data, where extreme tails, systematic missingness, and temporal structure distinguish the setting from generic tabular benchmarks.

**Financial distress prediction.** Machine learning for bankruptcy and distress prediction has a long history, from Altman [1968]'s Z-score to modern ensemble methods. Barboza et al.

[2017] benchmarked random forests and boosting against traditional statistics, establishing ML superiority. Mai et al. [2019] applied deep learning to textual disclosures alongside accounting ratios. More recently, Korangi et al. [2023] applied Transformers to structured financial panel data for corporate default prediction, and Chen et al. [2022] demonstrated the value of high-dimensional XBRL-tagged features over traditional small variable sets. Our work differs in using raw XBRL features extracted directly from SEC EDGAR (rather than pre-curated Compustat fields) and in validating self-supervised pretraining in this domain.

**Self-supervised pretraining for tabular data.** Adapting SSL to tabular data is a recent research direction. VIME [Yoon et al., 2020] introduced mask estimation and value imputation as pretext tasks for tabular features. SCARF [Bahri et al., 2022] applied contrastive learning via random feature corruption, while SubTab [Ucar et al., 2021] used feature subsetting to create multiple views. Our masked feature reconstruction objective is closest to VIME but uses a simpler masking strategy (zero-replacement rather than learned corruption) and reconstructs from a global CLS representation rather than per-feature outputs.

# 7 Open-Source Release

We release the following artifacts alongside this report:

1. **EDGAR XBRL extraction pipeline**—automated download and parsing of Company Facts API data for the full SEC filer universe.
2. **Feature definitions and normalization code**—16 raw features, 12 financial ratios, 5 YoY changes, with configurable winsorization and quantile normalization.
3. **Company universe and split specifications**—survivorship-bias-free universe construction and reproducible temporal split definitions.
4. **Training and evaluation scripts**—FT-Transformer, baseline models, SSL pretraining, and the full evaluation harness with all reported metrics.

Code repository: `https://github.com/elroy-galbraith/fin-jepa.git`

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